HCI FUNDAMENTALS 2026

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Low-Fi Prototypes: Exercises

Human Computer Interaction

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Exercise 1

- The following slides includes two paper prototypes from the 2022 edition of the course
  - **BEWARE: pre-feedback!**
  - **DO NOT** use them as examples of good prototypes!
- Work in groups or pairs
- Using the design principles discussed in class and the tasks reported for *each* prototype:
  - criticize each paper prototype
  - make a list of pros and cons for the design

Prototype 1: Math-ilo tu!

- Value proposition: *Turning Math into Reality*
- Goal: to support elementary school teachers teaching math
- Tasks:
  - Simple: Choose the most suitable way to represent and analyze a math problem
  - Moderate: Personalize the problem according to children's preferences for a more engaging analysis
  - Complex: Show different step-by-step resolutions of the math problem to facilitate students' comprehension
- Prototype: https://polito-hci-2023.github.io/materials/slides/05b-exercise/01a-math-ilo-tu-prototype.pdf
- Flow diagram: https://polito-hci-2023.github.io/materials/slides/05b-exercise/01b-math-ilo-tu-prototype-flow.pdf

Prototype 2: TouchGrass

- Value proposition: *Take a rest. Enjoy your meal!*
- Goal: to disconnect from work during mealtime
- Tasks:
  - Simple: Restrict work related phone usage at the beginning of a mealtime
  - Moderate: Earn rewards after disconnecting for some hours
  - Complex: Become a reward provider as a company
- Prototype: https://polito-hci-2023.github.io/materials/slides/05b-exercise/02a-touchgrass-prototype.pdf
- Flow diagram: https://polito-hci-2023.github.io/materials/slides/05b-exercise/02b-touchgrass-prototype-flow.pdf

Exercise 2

- Select <u>one</u> of the previous prototypes and 1-2 *tasks* from it
- Pick a different *modality* than the one adopted in the prototype
  - modality: a way of interacting or a device type
  - e.g., if the prototype is tablet + AR, pick mobile + touch or desktop + VR or…
- Design a few significant pages of the prototype to complete the tasks, starting from the main one
- Take a photo/scan and share it on Telegram (topic: “in-class exercises”)

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